1. A company has $3,000 and wants to buy a block of shares from the following companies. The company needs to select which companies to invest in with the objective of maximizing annual return on investment with respect to the following requirements:
   - At least two Texas companies must be in the portfolio.
   - No more than one investment in foreign companies.
   - Exactly one of two California companies must be selected.
   - If British Petro is included, then Texas Oil must be included as well.

Company Name (Location)	Expected Annual Return	Cost of Share
Texas Oil (Texas)	$50	$480
British Petro (foreign)	$80	$540
Dutch Shell (foreign)	$90	$680
Houston Drilling (Texas)	$120	$1,000
Lone Star Petro (Texas)	$110	$700
San Diego Oil (California)	$40	$510
Cal Petro (California)	$75	$900

$$Max\ 50T + 80B + 90D + 120H + 110L + 40S + 75C$$

S.t:

$$T + H + L \geq 2$$
$$B + D \leq 1$$
$$S + C = 1$$
$$B \leq T$$
$$480T + 540B + 680D + 1{,}000H + 700L + 510S + 900C \leq 3{,}000$$
$$T, B, D, H, L, S, C \in \{0,1\}$$

2. A software company, Star Trail, has identified several new software products. Each software product has a cost and requires human resource commitments. Estimated profits for each project have also been identified. All these estimates are tabulated below. Projects A and F will require the same programmer's expertise, so only one can be pursued. Projects B and E yield handsome profits but require substantial financial commitments and entail significant risks. Therefore, Star Trail must take one of the two projects. Star Trail can afford to invest at most $200,000 in new projects and wants to generate a profit of at least $400,000. Formulate the problem as an IP model, which minimizes the total number of programmers required.

Software	Cost	# Of programmers required	Estimated Profit
A	$35,000	3	$90,000
B	$100,000	6	$190,000
C	$50,000	5	$120,000
D	$40,000	5	$100,000
E	$120,000	9	$260,000
F	$20,000	3	$55,000

$$Min\ 3A + 6B + 5C + 5D + 9E + 3F$$

S.t:

$$35{,}000A + 100{,}000B + 50{,}000C + 40{,}000D + 120{,}000E + 20{,}000F \leq 200{,}000$$
$$90{,}000A + 190{,}000B + 120{,}000C + 100{,}000D + 260{,}000E + 55{,}000F \geq 400{,}000$$
$$A + F \leq 1$$
$$B + E = 1$$
$$A, B, C, D, E, F \in \{0,1\}$$